

Multiple Choice

1. **Answer:** A

Options: A) to and fro; B) in the direction opposite to the flow of current; C) in a circular direction; D) only when the circuit is closed; E) in a random direction

Note: Correct option: A - to and fro

2. **Answer:** E

Options: A) 7.1; B) 9.4; C) 4.0; D) 3.8; E) 5.33

Note: Correct option: E - 5.33

3. **Answer:** A

Options: A) Methane; B) Nitrogen; C) Oxygen; D) Iron; E) Hydrogen

Note: Correct option: A - Methane

4. **Answer:** E

Options: A) dispersion; B) deflection; C) refraction; D) reflection; E) interference

Note: Correct option: E - interference

5. **Answer:** A

Options: A) 600 watts; B) 800 watts; C) 750 watts; D) 1000 watts; E) 700 watts

Note: Correct option: A - 600 watts

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Charge'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'increases at first, then decreases.'

Long Form Response

11. **Answer:** - 3480 N. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 5.8 ns. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key